REDHA NACEF

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TECHNICAL SKILLS

Languages and Technologies: Python, C++, C#, React, Javascript, Typescript, Arduino, Three.js

WORK EXPERIENCE

Software Engineer: Global Design Innovation 2022 - Present

Tech Stack: Python, C#, C++, Typescript/React, AutoCAD, AWS, Azure Devops, PostgreSQL, Github, PyTorch, Langchain, Three.js, Vue.js

- Worked across the full stack on GDi's Vision software, with a particular focus on the 3D viewer built using **Three.js** and **Vue.js**. Implemented new features and refactored legacy code, resulting in a **2× increase in FPS performance** and **a significant reduction in memory usage.**
- Worked in partnership with the University of Aberdeen to create pipe **modelling software** that processed laser scans from industrial assets and automatically modelled sections of pipelines using **computer vision and machine learning**, significantly reducing manual modeling time for piping engineers in **AutoCAD**.
- Led a proof-of-concept for the development of a web application that uses **generative AI** to aid in the completion of technical documents in heavy industry. This included researching and implementing emergent generative AI technologies, incorporating several **open source libraries**, managing code repositories with git, **building an AI testing program and framework** to ensure consistent and reliable results, and both front- and back-end development.

Process Engineer: Impact Recycling Oct 2020 - Feb 2022

Tech Stack: Python, Solidworks, AutoCAD, Visio

- Led and contributed to successful applications for Innovate UK grants worth **millions of pounds** for the company and managed multiple projects concurrently after achieving funding, **overseeing planning and execution of activities**.
- Led and carried out R&D projects **validating and scaling-up** new cutting-edge technologies and processes in plastic recycling, including the development of a novel plastic recycling process from the initial R&D stage, through to small-scale prototyping and eventual commissioning of a full-scale plant.

PROJECT EXPERIENCE

Painting Music 2023

Worked as an engineer in a small team for a project started at the University of Aberdeen in collaboration with an Aberdeen based artist exploring AI and creativity. Implemented an AI model using Python trained on different genres of music that produces new and spontaneous compositions in reaction to live art pieces as they are created.

Acoustic Levitation Summer 2019

Designed and built an ultrasonic levitation device over three months, coding microcontrollers with Arduino and Python to produce a device capable of levitating small spheres using only sound waves.

Thermophoresis of Fluid Mixtures in Micro-gravity Autumn 2019

Shortlisted for the European Space Agency Fly Your Thesis initiative for our proposed project investigating thermophoresis of fluid mixtures in microgravity.

EDUCATION

University of Aberdeen: MEng Chemical Engineering: (First Class Honours) 2015 - 2020

Master's Thesis: Ultrasonic Levitation Supervised by Dr Marcus Bannerman

Achievements : **Francis Morrison Prize in Engineering**
Carnegie Vacation Scholarship
James Main Prize in Engineering

ADDITIONAL INFORMATION

Driving Licence:	UK Cat B (Car)
Languages:	English, Arabic (Algerian)
Hobbies & Interests:	Sports (Football, Volleyball, Running), Trumpet, Programming, Game Development, 3D printing, Electronics, Reading, Dressmaking.